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Survey Paper

The Hopf algebra of linearly recursive sequences

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Abstract. We explain how the space of linearly recursive sequences over a field can be considered as a Hopf algebra. The algebra structure is that of divided-power sequences, so we concentrate on the perhaps lesser-known coalgebra (diagonalization) structure. Such a sequence satisfies a minimal recursive relation, whose solution space is the subcoalgebra generated by the sequence. We discuss possible bases for the solution space from the point of view of diagonalization. In particular, we give an algorithm for diagonalizing a sequence in terms of the basis of the coalgebra it generates formed by its images under the difference-operator shift. The computation involves inverting the Hankel matrix of the sequence. We stress the classical connection (say over the real or complex numbers) with formal power series and the theory of linear homogeneous ordinary differential equations. It is hoped that this exposition will encourage the use of Hopf algebraic ideas in the study of certain combinatorial areas of mathematics.

1. Introduction

A linearly recursive sequence over a field F is a sequence of elements of F such that beginning from a certain term, each term is a fixed linear combination of previous terms. It is well-known that these sequences arise in number theory, finite differences, differential equations and power series calculations, among other areas of mathematics. What seems not to be so well-known is that these sequences form a coalgebra over F, which is the so-called continuous dual to the algebra $F[x]$ of polynomials in one variable over F. In fact, these sequences form a Hopf algebra over F. Since it is becoming apparent that there are important connections between Hopf algebras and certain combinatorial areas of mathematics, it seems appropriate to discuss certain coalgebraic aspects of these sequences.

Since we do not wish to assume any special acquaintance with Hopf algebras, we include in section 2 a description of some of the basic ideas about coalgebras and Hopf algebras, with appropriate examples. A basic reference for these ideas is [5]. We do not always include formal proofs, since the main objective is to place these sequences in the proper perspective. We occasionally refer to the literature for a formal proof or further theoretical details, but in any case we hope to make clear the type of calculations one can do with these sequences.

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In section 3, we discuss algebras and coalgebras as dual notions, and identify the linearly recursive sequences as the appropriate dual to $F[x]$. In sections 4 and 5, we examine in detail the coalgebra structure of these sequences. Each such sequence satisfies a minimal recursive relation. The solution space of this relation is the subcoalgebra generated by the sequence. In section 4, we give an algorithm for diagonalizing a sequence in terms of the basis of the coalgebra it generates formed by its images under a shift operator. In section 5, we discuss the general question of bases for a solution space. Over the real numbers, we indicate the parallel with the theory of linear homogeneous differential equations. Throughout we have tried to interpret each notion for characteristic 0 in more familiar terms involving power series. The danger is that at non-zero characteristic, there is also a power-series interpretation, but it is one of so-called "divided" powers rather than "formal" power series.

The treatment is somewhat discursive, as we try to emphasize computational aspects over formal theory. The final section 6 mentions some other possible questions one could consider.

2. Hopf algebra notions

F will denote an arbitrary field. An algebra (associative algebra with unit element) is given by a vector space A over F, a multiplication map $m: A \otimes A \to A$ and a unit map $\mu: F \to A$. The associativity of m is expressed by the commutativity of the diagram

$$\begin{tikzcd} A \otimes A \otimes A \arrow[r, "m \otimes I"] \arrow[d, "I \otimes m"'] & A \otimes A \arrow[d, "m"] \\ A \otimes A \arrow[r, "m"] & A \end{tikzcd},$$

and the unit property is expressed by the commutativity of the diagram

$$\begin{tikzcd} F \otimes A \arrow[r, "\mu \otimes I"] & A \otimes A \arrow[d, "m"] & A \otimes F \arrow[l, "I \otimes \mu"'] \\ & A \arrow[ul, leftrightarrow] \arrow[ur, leftrightarrow] & \end{tikzcd},$$

where $\leftrightarrow$ is a canonical isomorphism. The unit element of A is $\mu(1)$, which by abuse of notation, is also written as 1.

We mention several examples of algebras. In this paper, we shall concentrate on the algebra $A = F[x]$ of polynomials in x. If M is a (multiplicative) monoid, then the monoid algebra $F[M]$ consists of all formal finite sums $\sum_{i=1}^{n} \alpha_i m_i$, $\alpha_i \in F$, $m_i \in M$, with multiplication given by $(\sum_i \alpha_i m_i)(\sum_j \beta_j m_j) = \sum_{i,j} \alpha_i \beta_j (m_i m_j)$. $F[x]$ is a special case, where M is the multiplicative monoid consisting of all non-negative powers of x.

For a multiplicative monoid M, the set Map (M, F) of all functions from M to F is an algebra under pointwise operations. The unit element is the constant function 1. For $a \in M$, let r_a and l_a denote right and left multiplication by a in M. A function $f \in \mathrm{Map}\,(M, F)$ is called a *representative* function if $\{f \circ r_a \mid a \in M\}$ has finite-dimensional F-span. See [1] and [2] for a discussion of these functions and for an explanation of remarks not proved here. The set $R(M)$ consisting of all representative functions of M to F is a subalgebra of Map (M, F). Examples of representative functions are constant functions $f = \alpha$, since $f \circ r_a = f$; $f \in \mathrm{Hom}\,(M, (F, \cdot))$ since $f \circ r_a = f(a)f$; and $f \in \mathrm{Hom}(M, (F, +))$, since $f \circ r_a = f + f(a)1$.

Another example of an algebra is $F[[y]]$, the algebra of formal power-series in y. We note that the algebras $F[x]$, $R(M)$ and $F[[y]]$ are commutative, i.e., if $T: A \otimes A \to A \otimes A$ is the "twist" map sending $a \otimes b$ to $b \otimes a$, then $m \circ T = m$. $F[M]$ is commutative if and only if M is commutative.

We give a final example of a commutative algebra, the algebra D of divided powers. This has a countably infinite basis $d_0 = 1$, d_1, d_2, $d_3, \ldots$ with multiplication m given by $d_n d_m = \binom{n+m}{n} d_{n+m}$, where $\binom{n+m}{n}$ is the usual binomial coefficient. If F has characteristic 0, then D is isomorphic as an algebra to $F[x]$, where d_n corresponds to $x^n/n!$. This accounts for the name "divided powers". If F has positive characteristic p, then D is not isomorphic to $F[x]$. In particular, D has non-zero zero-divisors, e.g., $d_1 d_{p-1} = 0$ whereas $F[x]$ is an integral domain.

The notion of a *coalgebra* (coassociative coalgebra with counit) is obtained by reversing the arrows in the diagrams defining an algebra. Thus we have a vector space C over F, a comultiplication (or diagonalization) $\Delta: C \to C \otimes C$ and a counit $\varepsilon: C \to F$, and commutative diagrams

$$\begin{tikzcd} C \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & C \otimes C \arrow[d, "\Delta \otimes I"] \\ C \otimes C \arrow[r, "I \otimes \Delta"] & C \otimes C \otimes C \end{tikzcd}$$

and

$$\begin{tikzcd} & C \arrow[d, "\Delta"] \arrow[dl, leftrightarrow] \arrow[dr, leftrightarrow] & \\ F \otimes C & C \otimes C \arrow[l, "\varepsilon \otimes I"'] \arrow[r, "I \otimes \varepsilon"] & C \otimes F. \end{tikzcd}$$

Thus, if $c \in C$ and $\Delta c = \sum_i c_i \otimes c_i'$, then $\sum_i (\Delta c_i) \otimes c_i' = \sum_i c_i \otimes (\Delta c_i')$ and $\sum_i \varepsilon(c_i)c_i' = c = \sum_i \varepsilon(c_i')c_i$).

We give some examples of coalgebras. First consider $F[x]$. If we set $\Delta x = 1 \otimes x + x \otimes 1$, then Δ extends to an algebra homomorphism Δ of $F[x]$ to $F[x] \otimes F[x]$. In particular, $\Delta(x^n) = \sum_{i=0}^n \binom{n}{i} x^i \otimes x^{n-i}$. It is easy to check that Δ is coassociative. Also, if we set $\varepsilon(x) = 0$, then ε extends to an algebra homomorphism of $F[x]$ to F, with $\varepsilon(x^n) = \delta_{0,n}$ (δ_{ij} is 1 when $i = j$ and 0 when $i \neq j$.) ε is a counit for $F[x]$. Next recall D, which has a basis $\{d_n\}_{n=0}^\infty$. We can give D a coalgebra structure by defining $\Delta d_n = \sum_{i=0}^n d_i \otimes d_{n-i}$, and $\varepsilon(d_n) = \delta_{0,n}$. D, as coalgebra, is called the coalgebra of divided powers. When F has characteristic 0, our isomorphism σ between D and $F[x]$ associating d_n and $x^n/n!$ is easily seen to be a coalgebra isomorphism, i.e., $(\sigma \otimes \sigma)\Delta_D = \Delta_{F[x]}\sigma$ and $\varepsilon_{F[x]}\sigma = \varepsilon_D$.

If S is any set, then $F[S]$, the F-vector space with basis S, is a coalgebra via $\Delta a = a \otimes a$ and $\varepsilon(a) = 1$ for $a \in S$. In any coalgebra C, an element $a \neq 0$ with $\Delta a = a \otimes a$ and (necessarily) $\varepsilon(a) = 1$ is called a *group-like* element of C. The set $G(C)$ of group-like elements of C is linearly independent ([5]).

Let M be a (multiplicative) monoid, $R(M)$ the representative functions on M. We indicate a coalgebra structure for $R(M)$. See [1] and [2] for details. One can show that $R(M) \otimes R(M)$ is isomorphic (as algebra) to $R(M \times M)$ via the identification $(f \otimes g)(a, b) = f(a)g(b)$. The diagonalization Δ of $R(M)$ is thus described by $\Delta f = \sum f_i \otimes g_i$ means $f(ab) = \sum f_i(a)g_i(b)$ for all $a, b \in M$. Also $\varepsilon(f) = f(e)$, e the unit element of M. For example, if $f \in \operatorname{Hom}(M, (F, \cdot))$, then $\Delta f = f \otimes f$, i.e., f group-like, and if $f \in \operatorname{Hom}(M, (F, +))$, then $\Delta f = 1 \otimes f + f \otimes 1$.

A coalgebra C is called *cocommutative* if $T\Delta = \Delta$ (T = twist). Note that $F[x]$, D and $F[M]$ are cocommutative coalgebras. $R(M)$ is a cocommutative coalgebra if M is a commutative monoid.

At this point, observe that we have four examples $F[x]$, D, $F[M]$ and $R(M)$, each of which is simultaneously an algebra and a coalgebra. The easiest way to tie the two structures together is to note that in each case Δ and ε are algebra homomorphisms, where $A \otimes A$ has the usual algebra structure defined by $(a \otimes b)(a' \otimes b') = (aa' \otimes bb')$ whenever A is an algebra.

A *bialgebra* B is vector space which is an algebra via m and μ, a coalgebra via Δ and ε and such that Δ and ε are algebra homomorphisms. Examples are $F[x]$, D, $F[M]$ and $R(M)$. We note that if F has characteristic 0, then $F[x]$ and D are isomorphic as bialgebras.

A bialgebra H is called a *Hopf algebra* if it possesses an *antipode* S. S is a linear map of H to H satisfying the following property: if $h \in H$ and $\Delta h = \sum_i h_i \otimes h'$, then $\sum_i S(h_i)h_i' = \varepsilon(h)1 = \sum_i h_i S(h_i')$. Such an S is necessarily unique and is both an algebra and coalgebra anti-endomorphism of H (see [5] for details concerning S). If H is either commutative or cocommutative, then $S^2 = I$. If $a \in G(H)$, then a is invertible with $a^{-1} = S(a)$ and $G(H)$ is a group.

We give some examples of Hopf algebras. $F[x]$ is a Hopf algebra with $S(x^n)=(-1)^n x^n$. D is a Hopf algebra with $S(d_n)=(-1)^n d_n$. If F has characteristic 0, then $F[x]$ and D are isomorphic as Hopf algebras (i.e., the isomorphism preserves the action of the antipodes). Let G be a (multiplicative) group. Then $F[G]$ and $R(G)$ are Hopf algebras. For $F[G]$, $S(g)=g^{-1}$ for $g\in G$. For $f\in R(G)$, $(Sf)(g)=f(g^{-1})$ for $g\in G$.

Finally, we note that we have seen examples of bialgebras B with elements p satisfying $\Delta p = 1\otimes p + p\otimes 1$, e.g., the element x in $F[x]$. Such an element is called a *primitive* element of B. The set $P(B)$ of primitive elements of B is a Lie algebra, i.e., a subspace closed under $[p, q]=pq-qp$. For a primitive element p, note that $\varepsilon(p)=0$ and $S(p)=-p$ if B has an antipode S. As another example of a primitive element, if F has characteristic $p>0$, then x^p is a primitive element in $F[x]$. In fact, when F has characteristic $p>0$, then $P(B)$ is a p-Lie algebra, i.e., also closed under p-th powers.

3. Duality

Let C be a coalgebra. Let $C^* = \mathrm{Hom}\,(C, F)$, the full linear dual of C. We can consider C^* as an algebra with unit element ε, the counit of C. The multiplication in C^* is called convolution, and will be denoted by $*$. It is defined as follows: for f, $g\in C^*$, $c\in C$, $(f*g)(c)=(f\otimes g)\,\Delta c$, i.e., if $\Delta c=\sum_i c_i\otimes c_i'$, then $(f*g)(c)=\sum_i f(c_i)g(c_i')$. The coassociativity of Δ implies that $*$ is associative. C^* is called the *dual algebra* of C. C^* is commutative if and only if C is cocommutative.

We describe several examples. First consider D, the coalgebra of divided powers. Since a linear function on D is uniquely determined by its values on the basis $\{d_n\}_{n=0}^{\infty}$, we may identify elements of D^* with sequences $\{f_n\}_{n=0}^{\infty}$ of scalars from F, or with power-series $\sum_{n=0}^{\infty} f_n y^n$ in $F[[y]]$. The latter identification turns out to be appropriate, in the sense that it gives an algebra isomorphism between D^* and $F[[y]]$. The basic calculation to see this is as follows: y^i stands for the function $y^i(d_j)=\delta_{ij}$. Then

$$\begin{aligned}(y^i*y^j)(d_n)=(y^i\otimes y^j)\Delta d_n &= \sum_{k=0}^{n} y^i(d_k)\otimes y^j(d_{n-k})\\ &= \sum_{k=0}^{n} \delta_{ik}\delta_{j,n-k} = \begin{Bmatrix} 0 & \text{if} & i+j\neq n\\ 1 & \text{if} & i+j=n\end{Bmatrix} = \delta_{i+j,n} = y^{i+j}(d_n),\end{aligned}$$

i.e., $y^i*y^j=y^{i+j}$. Since $y^0(d_n)=\delta_{0n}=\varepsilon(d_n)$, we see that D^* is $F[[y]]$ as an algebra. The above calculation is sufficient since a power-series in $F[[y]]$ acts as a polynomial on a given element of D. If characteristic $F=0$, then $D\cong F[x]$, so we get $F[x]^*\cong F[[y]]$, where $y^i(x^j)=\delta_{ij}i!$. In general, for $C=F[x]$, if we identify C^*

with $F[[Z]]$, where $Z^i(x^j)=\delta_{ij}$, then a calculation similar to the one above shows that $Z^i * Z^j = \binom{i+j}{i} Z^{i+j}$, so that we can think of $F[x]^*$ as the algebra of divided power series.

For a set S, if $C=F[S]$, then $C^*=\mathrm{Map}(S, F)$ with pointwise product. For if f, $g \in C^*$, $s \in S$, then $(f*g)(s)=(f\otimes g)(\Delta s)=(f\otimes g)(s\otimes s)=f(s)g(s)$.

The passage from algebras to coalgebras is more complicated, in general, although it will be accessible for our basic example $A=F[x]$. The difficulty is that whereas the multiplication in C^* is obtained by dualizing Δ. i.e., putting together $\Delta^*:(C\otimes C)^* \to C^*$ with the natural map $C^*\otimes C^* \to (C\otimes C)^*$, the dualization m^* of the multiplication m gives $m^*: A^* \to (A\otimes A)^*$, and the image may not lie in $A^*\otimes A^*$. (This problem would not arise if A were finite-dimensional). Hence we must restrict ourself to an appropriate subspace of A^*. It will be denoted by A°. We shall give two descriptions of A°. See [5] and [6] for more details concerning these two approaches.

We first describe A° in a way which will enable us to realize $F[x]^\circ$ as linearly recursive sequences. A° is defined as $\{f\in A^* \mid \text{kernel } f \supset \text{a cofinite ideal } J \text{ of } A\}$. The conditions means $f(J)=0$ for some ideal J of A such that A/J is finite-dimensional over F. Then it turns out that $A^*\otimes A^* \to (A\otimes A)^*$, the injection given by $(f\otimes g)(a\otimes b)=f(a)g(b)$, when restricted to $A^\circ\otimes A^\circ$, yields an isomorphism of $A^\circ\otimes A^\circ$ with $(A\otimes A)^\circ$. Then m^*, restricted to A°, yields a diagonalization $\Delta: A^\circ \to A^\circ\otimes A^\circ$, which is described by $\Delta f=\sum f_i\otimes g_i$ means $f(ab)=\sum f_i(a)g_i(b)$ for a, $b\in A$. The counit ε of A° is evaluation at the unit element of A, i.e., $\varepsilon(f)=f(1)$. A° is a coalgebra, called the *continuous dual* of the algebra A. A° consists of the continuous linear functions in A^* where F is considered as a discrete space and A is a topological group with a neighborhood base at 0 consisting of the cofinite ideals. A° will be cocommutative if A is commutative.

The preceding equations are reminiscent of the coalgebra description of $R(M)$, the representative functions on a monoid M. In fact, if one chooses M as the multiplicative monoid of A, then A° is precisely $A^*\cap R(M)$, the linear representative functions on A, i.e., all linear functions on A with finite-dimensional right translates. From this point of view, A° is a subcoalgebra of $R(M)$. Another remark tying these ideas together is the following: For an arbitrary monoid M, if we identify $F[M]^*$ with $\mathrm{Map}(M, F)$, then $F[M]^\circ = R(M)$ as a coalgebra.

Now we turn to our chief example $A=F[x]$. We identify A^* with the space of all infinite sequences $f=(f_n)_{n=0}^\infty$, where $f(x^n)=f_n$. Let $f\in A^\circ$. Since all non-zero ideals J of $F[x]$ are generated by monic polynomials, they are all cofinite. Let $f(J)=0$ for such a non-zero J containing a monic polynomial

$$h(x)=x^r-h_1x^{r-1}-\cdots-h_r. \tag{1}$$

Then $f(x^m h(x))=0$ for $m\geq 0$, i.e.,

$$f_n=h_1f_{n-1}+h_2f_{n-2}+\cdots+h_rf_{n-r}\quad\text{for}\quad n\geq r. \tag{2}$$

A sequence $f=\{f_n\}_{n=0}^{\infty}$ is called *linearly recursive* if (2) holds for some $r>0$ and $h_1,h_2,\ldots,h_r\in F$. Conversely, if $f=\{f_n\}_{n=0}^{\infty}$ is linearly recursive, then $f(J)=0$ for the ideal J generated by $h(x)$ of (1). Hence A° consists of all linearly recursive sequences.

In the next section, we shall discuss the coalgebra structure of $F[x]^\circ$ in more computational detail. We conclude this section with some miscellaneous remarks, which will be useful in understanding $F[x]^\circ$.

If B is a bialgebra, then we have seen that B° is a coalgebra and B^* is an algebra. But also $B^\circ\subset B^*$, and B° is a subalgebra of B^*. In fact, B° is itself a bialgebra. If H is a Hopf algebra, then H° is a Hopf algebra, whose antipode S° is the restriction to H° of the transpose S^* of the antipode S of H, i.e., for $f\in H^\circ$, $a\in H$, we have $(S^\circ f)(a)=f(S(a))$.

We apply these remarks to $H=F[x]$. We have seen that H° is the coalgebra of linearly recursive sequences $f=\{f_n\}_{n=0}^{\infty}$. We identified H^* with the algebra of divided power series $F[[Z]]$, where $Z^i(x^j)=\delta_{ij}$, and $Z^iZ^j=\binom{i+j}{i}Z^{i+j}$. Here $\{f_n\}_{n=0}^{\infty}$ in H° is identified with $\sum_{n=0}^{\infty}f_nZ^n$. Thus the product of $\{f_n\}_{n=0}^{\infty}$ and $\{g_n\}_{n=0}^{\infty}$ in H° is $\{h_n\}_{n=0}^{\infty}$ where $h_n=\sum_{i=0}^{n}\binom{n}{i}f_ig_{n-i}$. The unit element of H° is the counit of H, i.e., $\varepsilon=\{\varepsilon_n\}_{n=0}^{\infty}$, where $\varepsilon_n=\delta_{0,n}$. The antipode of H° sends the sequence $\{f_n\}_{n=0}^{\infty}$ to $\{(-1)^nf_n\}_{n=0}^{\infty}$. Note that H° contains D in the sense that $d_n\to Z^n$ defines an algebra isomorphism of D into H°. Note that this is actually a Hopf algebra isomorphism of D into H°. To see this, we must show $\Delta Z^n=\sum_{i=0}^{n}Z^i\otimes Z^{n-i}$ in H°. To see this, apply both sides to $x^j\otimes x^k$. The result follows from $Z^n(x^{j+k})=\sum_{i=0}^{n}Z^i(x^j)Z^{n-i}(x^k)$, both sides being $\delta_{n,j+k}$. Also $\varepsilon(Z^n)=Z^n(1)=\delta_{n,0}=\varepsilon(d_n)$, and $S(Z^n)=(-1)^nZ^n$ and $S(d_n)=(-1)^nd_n$. Thus D corresponds to the *finite* sequences in H°, i.e., those which become zero beyond some term. If the characteristic of F is zero, then we have seen that if we set $y^n=n!Z^n$, i.e., $y^i(x^j)=\delta_{ij}i!$, then H^* is isomorphic to the algebra of power series $F[[y]]$, so that we may view H° as a subalgebra of $F[[y]]$. In this case, D corresponds to the polynomials in $F[y]$.

Finally, we mention the so-called fundamental theorem of coalgebras. If C is a coalgebra, then this theorem asserts that every element of C lies in a finite-dimensional subcoalgebra of C. See [5] for a proof. For our purposes, we wish to discuss the coalgebra $R(M)$ of representative functions on a monoid M. If $f\in R(M)$, then the space C_f which is the span of all translates $\{f\circ r_a\circ l_b \mid a,b\in M\}$ is the subcoalgebra generated by f, i.e., the smallest subcoalgebra containing f, and it is finite-dimensional. If M is commutative, then C_f is the span of

$\{f \circ r_a \mid a \in M\}$, which is finite-dimensional by definition of representative function. Now if A is an algebra, we have seen that A° is a subcoalgebra of $R[M]$, where M is the multiplicative monoid A. Thus if $f \in A^\circ$, then the subcoalgebra of A° generated by f is C_f, the span of $\{f \circ r_a \circ l_b \mid a, b \in A\}$.

We apply this discussion to $A = F[x]$, where we have identified A° to the space of linearly recursive sequences $f = \{f_n\}_{n=0}^\infty$ via $f(x^n) = f_n$. Then C_f is spanned by $\{f \circ r_{x^n} \mid n = 0, 1, 2, \ldots\}$. But what is $f \circ r_x$? $(f \circ r_x)(x^n) = f(x^{n+1}) = f_{n+1}$, i.e., if $f = \{f_n\}_{n=0}^\infty$, then $f \circ r_x = \{f_{n+1}\}_{n=0}^\infty$. In terms of sequences $f = (f_1, f_1, f_2, \ldots)$, $f \circ r_x = (f_1, f_2, f_3, \ldots)$, i.e., $f \circ r_x = Df$, D the so-called difference-operator shift. More generally, $f \circ r_{x^n} = D^n f$. Thus C_f is spanned by $\{D^n f\}_{n=0}^\infty$. If f satisfies the recursive relation (2), then f, Df, $\ldots, D^{r-1}f$ span C_f, so C_f is finite-dimensional of dimension $\leq r$. We note that D is a derivation of the algebra $F[X]^\circ$. This follows from $\Delta x = 1 \otimes x + x \otimes 1$.

If characteristic $F = 0$ and we identify $f = \{f_n\}_{n=0}^\infty$ with

$$\sum_{n=0}^\infty f_n Z^n = \sum_{n=0}^\infty f_n \frac{y^n}{n!},$$

then $Df = \{f_{n+1}\}_{n=0}^\infty$ is identified with

$$\sum_{n=0}^\infty f_{n+1} \frac{y^n}{n!} = \sum_{n=1}^\infty f_n \frac{y^{n-1}}{(n-1)!},$$

so that $Df = df/dy$, i.e., D is formal differentiation with respect to y. Thus, if h and f are as in (1) and (2), then f is a formal solution of the differential equation

$$\frac{d^r\varphi}{dy^r} - h_1 \frac{d^{r-1}\varphi}{dy^{r-1}} - \cdots - h_r\varphi = 0$$

with initial conditions $\varphi^{(n)}(0) = f_n$ for $0 \leq n \leq r-1$.

4. Diagonalizing linearly recursive sequences: the coalgebra structure of $F[X]^\circ$.

We begin by identifying the group-like elements of H°, $H = F[x]$. For $f = \{f_n\}_{n=0}^\infty \in H^\circ$, $\Delta f = f \otimes f$ means $f(x^i)f(x^j) = f(x^{i+j})$, all $i, j \geq 0$, i.e., f is an algebra homomorphism of $F[x]$ to F. Thus $f_0 = 1$ and $f_n = f_1^n$ for $n \geq 1$, so f is completely determined by $f_1 = f(x)$. In our identification of H^* with the divided power series algebra $F[[Z]]$, f is the geometric series $\sum_{i=0}^\infty (f_1 Z)^i$. f vanishes on the ideal generated by $x - f_1$, and $C_f = Ff$. To identify $G(H^\circ)$ as a group, let $f = \{f_1^i\}_{i=0}^\infty$ and

$g=\{g_1^i\}_{i=0}^{\infty}$ be in $G(H^\circ)$. Then

$$(fg)(x)=(f\otimes g)(1\otimes x+x\otimes 1)=f(x)+g(x)=f_1+g_1,$$

i.e., $fg=\{(f_1+g_1)^i\}_{i=0}^{\infty}$. Hence $G(H^\circ)$ is isomorphic to the additive group of F. If characteristic F is 0, and we consider H° as a subalgebra of the power series algebra $F[[y]]$, then $\sum_{i=0}^{\infty}(f_1Z)^i=\sum_{i=0}^{\infty}f_1^iZ^i$ is replaced by $\sum_{i=0}^{\infty}f_1^i\, y^i/i!$, the exponential series for e^{f_1y}. Then $e^{f_1y}e^{g_1y}=e^{(f_1+g_1)y}$ makes it clear that $G(H^\circ)\cong(F,+)$.

Next we turn to the primitive elements of H°, i.e., elements $f=\{f_n\}_{n=0}^{\infty}$ such that $\Delta f=\varepsilon\otimes f+f\otimes\varepsilon$. This means $f(x^{i+j})=\delta_{0i}f(x^j)+\delta_{0j}f(x^i)$ for all i, $j\geq 0$. For $i=j=0$, this gives $f_0=0$. For $n\geq 2$, set $i=1$, $j=n-1$ to get $f_n=0$. Thus $f=f_1Z$, i.e., $f(x^n)=\delta_{n,1}f_1$. If $f\neq 0$, then f vanishes on the ideal generated by x^2, and $C_f=Ff+F\varepsilon$.

Somewhat more generally, we determine the $e(a)$-primitive elements f of H° for $a\in F$, i.e., $\Delta f=e(a)\otimes f+f\otimes e(a)$, where $e(a)$ is the group-like element $\{a^n\}_{n=0}^{\infty}=\sum_{n=0}^{\infty}a^nZ^n$. This means $f(x^{i+j})=a^if(x^j)+a^jf(x^i)$ for i, $j\geq 0$. Note that $e(a)Z=\sum_{n=0}^{\infty}(n+1)a^nZ^{n+1}$ is such, since

$$\Delta(e(a)Z)=(e(a)\otimes e(a))(\varepsilon\otimes Z+Z\otimes\varepsilon)=e(a)\otimes e(a)Z+e(a)Z\otimes e(a).$$

Conversely, if f is $e(a)$-primitive, then putting $i=j=0$ above shows $f_0=0$. Putting $i=j=1$ shows $f_2=2af_1$. Putting $i=1$, $j=n$ and arguing inductively, we get $f_{n+1}=(n+1)a^nf_n$, i.e., $f=f_1e(a)Z$. Thus the $e(a)$-primitives are precisely $Fe(a)Z$.

We recall that in section 3, we showed that $\{Z^n\}_{n=0}^{\infty}$ is a sequence of divided powers in H°, I.e., $\Delta Z^n=\sum_{i=0}^{n}Z^i\otimes Z^{n-i}$, $\varepsilon(Z^n)=\delta_{0,n}$. We can generalize this somewhat by noting that for $a\in F$, $\{e(a)Z^n\}_{n=1}^{\infty}$ is a sequence of divided powers over $e(a)$, i.e., $\Delta(e(a)Z^n)=\sum_{i=0}^{n}e(a)Z^i\otimes e(a)Z^{n-i}$.

Let $(H^\circ)^1$ be the F-span of $\{Z^n\}_{n=0}^{\infty}$. $(H^\circ)^1$ is a cocommutative irreducible (see [5]) coalgebra, isomorphic to D, called the irreducible component of 1 in H°. By Theorem 8.1.5 of [5], $H^\circ=F[G(H^\circ)]\otimes H^1$ as an algebra, using that H° is commutative. See also Theorems 13.0.1 and 13.2.2 of [5]. The effect of this is that H° has a basis $\{e(a)Z^n \mid a\in F,\ n\geq 0\}$. Recall that $e(a)Z^n\cdot e(b)Z^m=\binom{m+n}{n}e(a+b)Z^{m+n}$.

We now turn to the coalgebra structure of H°. Since $\{e(a)Z^n \mid a\in F, n\geq 0\}$ is a basis for H°, and for a fixed $a\in F$, $\{e(a)Z^n\}_{n=0}^{\infty}$ is a divided-power sequence over $e(a)$, we have a global description of Δ and ε. We will return to this point of view later.

We now discuss a "local" point of view, i.e., given an $f\in H^\circ$, describe Δf in terms of C_f, the coalgebra generated by f. Note that if $f=\{f_n\}_{n=0}^{\infty}$, then $\varepsilon(f)=f(1)=f_0$, so ε has an easy description.

Let $f=\{f_n\}_{n=0}^{\infty}$ be a non-zero linearly recursive sequence in H°. Recall that there is a polynomial

$$h(x)=x^r-h_1x^{r-1}-\cdots-h_r,\ r\geq 1 \tag{1}$$

such that

$$f_n=h_1f_{n-1}+h_2f_{n-2}+\cdots+h_rf_{n-r} \quad \text{for} \quad n\geq r. \tag{2}$$

We define two ideals I_f and J_f of $F[x]$. Let $I_f=\{h(x)\mid h(x)=0$ or $h(x)=h_0(x^r-h_1x^{r-1}-\cdots-h_r)$, $r\geq 1$, $h_0\neq 0$ and (2) holds for $n\geq r\}$. Let $J_f=\{h(x)\mid h(x)=0$ or $h(x)=h_0(x^r-h_1x^{r-1}-\cdots-h_r)$, $h_0\neq 0$ and (2) holds for n sufficiently large, i.e., $n\geq N$, with N perhaps larger than $r\}$. Since (2) holds for $n\geq$ some fixed N, it is clear that I_f and J_f are ideals of $F[x]$. Let $h_f(x)$ and $g_f(x)$ be monic generators of I_f and J_f respectively. Since $I_f\subset J_f$, we have that $g_f(x)\mid h_f(x)$. Let $g_f(x)=x^s-g_1x^{s-1}-\cdots-g_s$. Then $g_s\neq 0$. For if $g_s=0$, then $f_n=g_1f_{n-1}+\cdots+g_{s-1}f_{n-s+1}$ for n sufficiently large, i.e., $x^{s-1}-g_1x^{s-2}-\cdots-g_{s-1}$ is in J_f. Now if $f_n=g_1f_{n-1}+\cdots+g_sf_{n-s}$ for $n\geq N$, then $x^{N-s}g_f(x)=x^N-g_1x^{N-1}-\cdots-g_sx^{N-s}$ satisfies (2) for $n\geq N$, i.e., $x^{N-s}g_f(x)\in I_f$, so that $h_f(x)\mid x^{N-s}g_f(x)$. It follows easily that $h_f(x)=x^mg_f(x)$ for some m. Since $g_s\neq 0$, x^m is the largest power of x dividing $h_f(x)$. As an example, let $f=(1, 1, 1, 2, 3, 5, 8,\ldots)$, with $f_n=f_{n-1}+f_{n-2}$ for $n\geq 3$. Then $h_f(x)=x^3-x^2-x$ and $g_f(x)=x^2-x-1$.

A linearly recursive sequence $f=\{f_n\}_{n=0}^{\infty}$, is usually given by giving initial values $f_0, f_1,\ldots,f_{r-1}$ and a recursive relation (2). However, there is no guarantee that $h_f(x)$, the minimal recursive relation, is the same as $h(x)$. We give now a procedure for determining $h_f(x)$. We first compute the values $f_r, f_{r+1},\ldots,f_{2r-2}$. We then form the symmetric r by r matrix $H(f, h(x))$, indexed by $0\leq i, j\leq r-1$, whose ij-th entry is f_{i+j}. Thus the i-th row $f^{(i)}$ of $H(f, h(x))$ is $(f_i, f_{i+1},\ldots,f_{i+r-1})$. $H(f, h(x))$ is called the *Hankel matrix of f and $h(x)$*. Let t be the largest integer $(\leq r)$ such that $f^{(0)}, f^{(1)},\ldots,f^{(t-1)}$ are linearly independent rows. If $t=r$, then $h_f(x)=h(x)$ as in (1). If $t<r$, then $f^{(t)}=h_1'f^{(t-1)}+\cdots+h_t'f^{(0)}$ for uniquely determined scalars $h_1', h_2',\ldots,h_t'$. This says that $f_n=h_1'f_{n-1}+\cdots+h_t'f_{n-t}$ for $t\leq n\leq t+r-1$. To see that this holds for all $n\geq t$, consider the sequence $g=\{g_i\}_{i=0}^{\infty}$, where $g_i=f_{i+t}-h_1'f_{i+t-1}-\cdots-h_t'f_i$. Note $g_0=g_1=\cdots=g_{r-1}=0$. But g satisfies the recursive relation obtained from $h(x)$, since f does. Hence $g=0$, and f satisfies the recursive relation obtained from $x^t-h_1'h_1'x^{t-1}-\cdots-h_t'$. Clearly f does not satisfy a recursive relation of degree $<t$, for then $f^{(0)}, f^{(1)},\ldots,f^{(t-1)}$ would be linearly dependent. Thus $h_f(x)=x^t-h_1'x^{t-1}-\cdots-h_t'$.

We call $h_f(x)$ the minimal recursive relation of f. Relabelling so that its degree is called r, we call r the minimal recursive degree of f. Note that r is the codimension in $F[x]$ of the ideal $I_f = (h_f(x))$. The Hankel matrix $H(f, h_f(x))$ will be called the *Hankel matrix of f* and denoted $H(f)$. The above discussion shows that $H(f)$ is a symmetric invertible matrix. The i-th row of $H(f)$ consists of the first r entries of $D^i f$ (recall $Df = \{f_{n+1}\}_{n=0}^{\infty}$, and that $f, Df, \ldots, D^{r-1}f$ span C_f). By our discussion of $H(f)$, we see that $f, Df, \ldots, D^{r-1}f$ are linearly independent, so that C_f has dimension r, with basis $f, Df, \ldots, D^{r-1}f$.

We will discuss Δf in terms of this basis $f, Df, \ldots, D^{r-1}f$. Since C_f is a coalgebra, it follows that for $0 \le k \le r-1$,

$$\Delta(D^k f) = \sum_{i,j=0}^{r-1} \alpha_{ijk}(D^i f) \otimes (D^j f) \tag{3}$$

for scalars $\alpha_{ijk} \in F$. The problem is to determine the α_{ijk}. In particular, taking $k=0$, we will have Δf.

Let $H(f)^{-1} = (s_{ij})$. Note that $s_{ij} = s_{ji}$ since $H(f)$ is symmetric. Thus we have the identity

$$\sum_{l=0}^{r-1} f_{l+m} s_{l,i} = \delta_{m,i} \quad \text{for} \quad 0 \le i, m \le r-1. \tag{*}$$

We introduce polynomials $s_i(x)$ for $0 \le i \le r-1$ by $s_i(x) = \sum_{j=0}^{r-1} s_{ji} x^j$. Then for $0 \le i,\ m \le r-1$,

$$(D^m f)(s_i(x)) = \{f_{l+m}\}_{l=0}^{\infty}\left(\sum_{j=0}^{r-1} s_{ji} x^j\right) = \sum_{l=0}^{r-1} f_{l+m} s_{l,i} = \delta_{m,i}.$$

We apply (3) to $s_i(x) \otimes s_j(x)$, $0 \le i,\ j \le r-1$. The right side of (3) yields

$$\sum_{l,m=0}^{r-1} \alpha_{l,m,k}((D^l f)(s_i(x)))((D^m f)(s_j(x))) = \sum_{l,m=0}^{r-1} \alpha_{l,m,k} \delta_{li} \delta_{mj} = \alpha_{i,j,k}.$$

The left side of (3) yields

$$(D^k f)(S_i(x) S_j(x) = (D^k f)\left(\sum_{l,m=0}^{r-1} S_{l,i} S_{m,j} x^{l+m}\right) = \sum_{l,m=0}^{r-1} f_{l+m+k} S_{l,i} S_{m,j}.$$

Hence, we have

$$\alpha_{ijk} = \sum_{l,m=0}^{r-1} f_{l+m+k} s_{l,i} s_{m,j}. \tag{4}$$

To compute these for $0 \le k \le r-1$, we must compute the terms of f through f_{3r-3}. However, for $k=0$, the terms through f_{2r-2} suffice, and they appear in $H(f)$. Using (4), we have that

$$\alpha_{ij0} = \sum_{m=0}^{r-1} \left(\sum_{l=0}^{r-1} f_{l+m} s_{li} \right) s_{mj} = \sum_{m=0}^{r-1} \delta_{mi} s_{mj} \text{ by (*), so that } \alpha_{ijo} = s_{ij}.$$

Thus (4) specializes to

$$\Delta f = \sum_{i,j=0}^{r-1} s_{ij} (D^i f) \otimes (D^j f). \tag{5}$$

As an example, let f be the Fibonacci sequence with $f_0 = f_1 = 1$ and $f_n = f_{n-1} + f_{n-2}$ for $n \ge 2$. Then $h_f(x) = x^2 - x - 1$ and $H_f(x) = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$. $H_f(x)^{-1} = \left[\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right]$, so that $\Delta f = 2(f \otimes f) - f \otimes Df - Df \otimes f + Df \otimes Df$. Another example, let f be the sequence discussed earlier with $f_0 = f_1 = f_2 = 1$ and $f_n = f_{n-1} + f_{n-2}$ for $n \ge 3$. Then $h_f(x) = x^3 - x^2 - x$ and

$$H(f) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} \text{ whose inverse is } \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

Thus

$$\Delta f = f \otimes f + f \otimes Df + Df \otimes f - f \otimes D^2 f - D^2 f \otimes f - 2(Df \otimes Df) + Df \otimes D^2 f + D^2 f \otimes Df.$$

5. The solution space of a linearly recursive relation

Given a non-zero linearly recursive sequence $f = \{f_n\}_{n=0}^{\infty}$, we have seen how to find the minimal recursive relation $h_f(x)$ and the ideal $I_f = (h_f(x))$. If degree $h_f(x) = r$, then I_f is of codimension r in $F[x]$, and the coalgebra C_f generated by f has dimension r. Let $I_f^{\perp} = \{g \in F[x]^{\circ} \mid g(I_f) = 0\}$. $I_f^{\perp}$ is a subcoalgebra of $F[x]^{\circ}$ (see [6]) which contains f. Hence $C_f \subseteq I_f^{\perp}$. But $I_f^{\perp}$ has dimension r, since one can prescribe the first r terms of an element f in $I_f^{\perp}$ arbitrarily, and then the remaining terms are determined. Hence $C_f = I_f^{\perp}$. Thus C_f can be described as the set of all linearly recursive sequences determined by the minimal recursive relation $h_f(x)$ of f.

We now consider the reverse point of view. We start with a monic polynomial $h(x)$ of degree $r \ge 1$. Let $I = (h(x))$ and $S_{h(x)} = I^{\perp} = \{f \in F[x]^{\circ} \mid f(I) = 0\}$. $S_{h(x)}$ is a subcoalgebra of $F[x]^{\circ}$, called the *solution space* of $h(x)$.

We first show how the prime power factorization of $h(x)$ splits $S_{h(x)}$ into a direct sum of subcoalgebras. Suppose $h(x)=p(x)q(x)$ where $p(x)$ and $q(x)$ are relatively prime. Then $I=(h(x))=(p(x))\cap(q(x))$. We claim that $I^{\perp}=(p(x))^{\perp}\oplus(q(x))^{\perp}$, i.e., $S_{h(x)}=S_{p(x)}\oplus S_{q(x)}$. $p(x)\mid h(x)$ means $(h(x))\subseteq(p(x))$ so that $S_{p(x)}\subseteq S_{h(x)}$. Similarly, $S_{q(x)}\subseteq S_{h(x)}$. Let $f\in(p(x))^{\perp}\cap(q(x))^{\perp}$. Since there exist polynomials $u(x)$ and $v(x)$ such that $u(x)p(x)+v(x)q(x)=1$, f vanishes on the ideal generated by 1, i.e., $f=0$. Hence $(p(x))^{\perp}\oplus(q(x))^{\perp}\subseteq I^{\perp}$. But if degree $p(x)=s$ and degree $q(x)=t$ with $s+t=r$, then we have noted that $I^{\perp}$ has dimension r, $(p(x))^{\perp}$ has dimension s and $(q(x))^{\perp}$ has dimension t. Hence $I^{\perp}=(p(x))^{\perp}\oplus(q(x))^{\perp}$.

We factor $h(x)=p_1(x)^{n_1}\cdots p_m(x)^{n_m}$ where $p_1(x),\ldots,p_m(x)$ are distinct primes, $n_1,\ldots,n_m$ are positive integers. Let $q_i(x)=p_i(x)^{n_i}$ for $1\le i\le m$. Then $S_{h(x)}=S_{q_1(x)}\oplus\cdots\oplus S_{q_m}(x)$. This is a direct sum of coalgebras.

We next consider $q(x)=(x-a)^n$ for $a\in F$. The $q_i(x)$ above will be of this form if F is algebraically closed, e.g., the complex numbers. We will show that $S_{q(x)}$ has a basis $e(a)$, $e(a)Z,\ldots,e(a)Z^{n-1}$, a (finite) sequence of divided powers over $e(a)$. Recall that $e(a)$ is the geometric series $\sum_{i=0}^{\infty}(aZ)^i$, Z^j is the linear function $Z^j(x^k)=\delta_{jk}$, and $e(a)Z^l$ is the convolution product of $e(a)$ and Z^l in $F[x]^{\circ}$. We saw in section 4 that $\{e(a)Z^i\}_{i=0}^{\infty}$ are linearly independent. Since $S_{q(x)}$ has dimension n, it suffices to show that $\{e(a)Z^i\}_{i=0}^{n-1}$ are in $S_{q(x)}$.

We proceed by induction. If $n=1$, then we must show that $e(a)((x-a)w(x))=0$ for any $w(x)\in F[x]$. But

$$e(a)((x-a)w(x))=\Delta e(a)((x-a)\otimes w(x))=(e(a)\otimes e(a))((x-a)\otimes w(x))=0$$

since $e(a)(x-a)=(1,a,a^2,\ldots)(x-a)=a-a=0$. Now suppose $n>1$, and that $\{e(a)Z^j\}_{j=0}^{n-2}$ are in $S_{(x-a)^{n-1}}$. Let $0\le i\le n-1$. Then

$$\begin{aligned}(e(a)Z^i)((x-a)^n w(x))&=\Delta(e(a)Z^i)((x-a)\otimes(x-a)^{n-1}w(x))=\\&=(e(a)\otimes e(a)Z^i)((x-a)\otimes(x-a)^{n-1}w(x))\\&\quad+\sum_{k=1}^{i}(e(a)Z^k\otimes e(a)Z^{i-k})((x-a)\otimes(x-a)^{n-1}w(x)).\end{aligned}$$

The first term is 0 since $e(a)(x-a)=0$. The remaining terms are 0 by induction, since $i-k\le i-1<n-1$ and $(e(a)Z^{i-k})((x-a)^{n-1}w(x))=0$.

If F has characteristic 0, then we have seen that replacing Z^i by $y^i/i!$ allows us to view the algebra H° as the power-series algebra $F[[y]]$. $e(ay)$ is replaced by e^{ay}, so that $S_{(x-a)^n}$ has a basis e^{ay}, $ye^{ay},\ldots,y^{n-1}e^{ay}$. In view of the theory of first order linear homogeneous differential equations, this is not surprising, since we have noted in section 3 that $S_{(x-a)^n}$ is the formal set of solutions to the differential equation whose characteristic equation is $(x-a)^n$.

Let F be the field of real numbers. We consider $q(x)=(x^2-bx-c)^n$, where x^2-bx-c is irreducible. Let $\alpha \pm \beta_i$ be the real roots of x^2-bx-c. Then we know from differential equations that $S_{q(x)}$ has a basis $e^{\alpha y}\cos \beta y$, $e^{\alpha y}\sin\beta y$, $ye^{\alpha y}\cos\beta y$, $ye^{\alpha y}\sin\beta y, \ldots, y^{n-1}e^{\alpha y}\cos\beta y$, $y^{n-1}e^{\alpha y}\sin\beta y$ in the power-series algebra $F[[y]]$. We can also consider the basis

$$\left\{\frac{y^i}{i!}e^{\alpha y}\cos\beta y\right\}_{i=0}^{n-1} \cup \left\{\frac{y^i}{i!}e^{\alpha y}\sin\beta y\right\}_{i=0}^{n-1}.$$

Since

$$\cos\beta y=\sum_{i=0}^{\infty}(-1)^i\frac{(\beta y)^{2i}}{(2i)!}=\sum_{i=0}^{\infty}(-1)^i\beta^{2i}\left(\frac{y^{2i}}{(2i)!}\right)$$

and

$$\sin\beta y=\sum_{j=0}^{\infty}(-1)^j\frac{(\beta y)^{2j+1}}{(2j+1)!}=\sum_{j=0}^{\infty}(-1)^j\beta^{2j+1}\left(\frac{y^{2j+1}}{(2j+1)!}\right),$$

we can consider $\cos\beta Z=\sum_{i=0}^{\infty}(-1)^i\beta^{2i}Z^{2i}$ and $\sin\beta Z=\sum_{j=0}^{\infty}(-1)^j\beta^{2j+1}Z^{2j+1}\in F[x]^\circ=F[[z]]$ as divided power algebra. Then $S_{q(x)}$ has a basis $e(\alpha)\cos\beta Z$, $e(\alpha)\sin\beta Z$, $Ze(\alpha)\cos\beta Z$, $Ze(\alpha)\sin\beta Z, \ldots, Z^{n-1}e(\alpha)\cos\beta Z$, $Z^{n-1}e(\alpha)\sin\beta Z$.

Let F be any field of characteristic 0, $h(x)\neq 0$ in $F[x]$. Suppose a basis of $S_{h(x)}$ is obtained in the power-series algebra $F[[y]]$, consisting of elements $\sum_{n=0}^{\infty}a_ny_n$. Then we can replace these by elements $\sum_{n=0}^{\infty}n!a_nZ^n$ to get a basis for $S_{h(x)}$ in $F[x]^\circ=F[[Z]]$. However, we cannot change from characteristic 0 to characteristic $p>0$, even when it seems to make sense. We illustrate this with two examples for $p=2$, one in which the change is harmless, and one in which it causes trouble. First let $h(x)=x^2+1$. At characteristic 0, a basis for $S_{h(x)}$ is $\{\cos y, \sin y\}$. Making the replacement in $F[[Z]]$, we get $\{\cos Z, \sin Z\}$, where $\cos Z=\sum_{i=0}^{\infty}(-1)^iZ^{2i}$ and $\sin Z=\sum_{i=0}^{\infty}(-1)^iZ^{2i+1}$. We can apply the technique of section 3 to diagonalize $\cos Z$ and $\sin Z$, noting that $D(\cos Z)=-\sin Z$ and $D(\sin Z)=\cos Z$. The Hankel matrix of $\cos Z$ is $\left[\begin{smallmatrix}1 & 0\\ 0 & -1\end{smallmatrix}\right]$, which is its own inverse, so $\Delta(\cos Z)=\cos Z\otimes\cos z-\sin Z\otimes\sin Z$. Similarly, $\Delta(\sin Z)=\cos Z\otimes\sin Z+\sin Z\otimes\cos Z$. Note that $\varepsilon(\cos Z)=1$ and $\varepsilon(\sin Z)=0$. S_{x^2+1} is known as the *trigonometric coalgebra.* Now let characteristic $F=2$. Then $x^2+1=(x+1)^2$ and so $S_{(x+1)^2}$ has a basis $e(1)$ and $e(1)Z$. Now $\cos Z=\sum_{i=0}^{\infty} Z^{2i}$ and $\sin Z=\sum_{i=0}^{\infty} Z^{2i+1}$. Hence $e(1)=\sum_{j=0}^{\infty} Z^j=\cos Z+\sin Z$. Note that $Z\sin Z=\sum_{i=0}^{\infty}\binom{2i+2}{1} Z^{2i+2}=0$ and $Z\cos Z=\sum_{i=0}^{\infty}\binom{2i+1}{1} Z^{2i+1}=\sin Z$. Hence $e(1)Z=\sin Z$. Hence the basis $\{\cos Z, \sin Z\}$ obtained from the characteristic 0 case, is still a basis for $p=2$.

Next let $h(x)=(x^2+1)^2$. At characteristic 0, $S_{h(x)}$ has a basis $\cos y$, $\sin y$, $y\cos y$, $y\sin y$. However, for $p=2$, these collapse to 2 elements $\cos Z$ and $\sin Z$, since $Z\cos Z=\sin Z$ and $Z\sin Z=0$. Where are 2 additional basis elements? At $p=2$, $h(x)=(x+1)^4$, so $S_{(x+1)^4}$ has a basis $e(1)$, $e(1)Z$, $e(1)Z^2$, $e(1)Z^3$. We saw that $e(1)=\sin Z+\cos Z$ and $e(1)Z=\sin Z$. Since $Z\cdot Z^2=Z^3$ when $p=2$, we have $e(1)Z^3=Z^2\sin Z$. Also $e(1)Z^2=Z^2\cos Z+Z^2\sin Z$. Hence another basis for $S_{(x+1)^4}$ when $p=2$ would be $\cos Z$, $\sin Z$, $Z^2\cos Z$, $Z^2\sin Z$. Of course the point is that x^2+1 is not irreducible when $p=2$. But if one used the basis of $S_{(x+1)^4}$ at characteristic 0 consisting of e^{-y}, ye^{-y}, y^2e^{-y}, y^3e^{-y}, then writing $y^2=2(y^2/2!)$ would give $2Z^2=0$ at $p=2$. So we would use instead e^{-y}, ye^{-y}, $y^2/2!\,e^{-y}$, $y^3/3!\,e^{-y}$, which would be replaced by $e(-1)$, $e(-1)Z$, $e(-1)Z^2$, $e(-1)Z^3$. This remains correct for $p=2$.

Hence if one is to utilize the theory of linear homogeneous differential equations of a real variable, one should always use solutions $(y^i/i!)e^{\alpha y}\cos\beta y$ and $(y^j/j!)e^{\alpha y}\sin\beta y$, which can be replaced by $Z^ie(\alpha)\cos\beta Z$ and $Z^je(\alpha)\sin\beta Z$ at arbitrary characteristic, assuming of course that α and β are in F.

We also note that the choice of basis for $S_{h(x)}$ can depend on the field F. For example, if $h(x)=x^2-x-1$, then we saw in section 4 that $S_{h(x)}$ has basis $f=(1, 1, 2, 3, \ldots)$ and $Df=(1, 2, 3, 5, \ldots)$ with

$$\Delta f=2(f\otimes f)-f\otimes Df-Df\otimes f+Df\otimes Df.$$

One can also compute by the equations (3) and (4) of section 4 that $\Delta(Df)=-(f\otimes f)+f\otimes Df+Df\otimes f$. This makes sense in any field F. One also has the "unit" bases $u_0=(1, 0, 1, 1, 2, 3, \ldots)$ and $u_1=(0, 1, 1, 2, 3, 5, \ldots)=Du_0$. Here one has that $\Delta u_0=u_0\otimes u_0+u_1\otimes u_1$ and $\Delta u_1=u_0\otimes u_1+u_1\otimes u_0+u_1\otimes u_1$, which also makes sense for any F. But if F is the real numbers, we may view $x^2-x-1=(x-\alpha)(x-\beta)$, where $\alpha=1+\sqrt{5}/2$ and $\beta=1-\sqrt{5}/2$. Then $S_{h(x)}$ has a basis of group-likes $e(\alpha)$ and $e(\beta)$ in $F[[Z]]$, or $e^{\alpha y}$ and $e^{\beta}y$ in $F[[y]]$. The entries of these sequences lie in the field $Q[\sqrt{5}]$ of rational numbers with $\sqrt{5}$ adjoined. If F is the rational numbers, then these group-likes do not lie in $F[x]^{\circ}$. If the characteristic of F is $p=5$, then $\alpha=\beta=3$ and the "real" method supplies only one basis element $e(3)$. A second basis element would be $e(3)Z$, as $x^2-x-1=(x-3)^2$ if $p=5$. If $p=2$, then the "real" method supplies no basis element at all.

6. Final remarks

In discussing a basis for C_f, we focused on f, $Df, \ldots, D^{r-1}f$, where r is the minimal recursive degree of f. Of course, $C_f=(h_f(x))^\perp$ also has the "unit" basis $u_0, u_1, \ldots, u_{r-1}$, where, for $0 \le i \le r-1$, u_i has a 1 in the i-th term, 0 in the j-th term for $j \ne i$, $0 \le j \le r-1$, and whose terms starting with the r-th term are determined by the recursive relation $h_f(x)$. Here the terms start with the 0-th term.) In general, one could look for a basis of $(h(x))^\perp$ whose diagonalization can be given by some algorithmic procedure, as we did for C_f.

Since $\{e(a)Z^i \mid a \in F,\ i \ge 0\}$ is a basis for $F[x]^\circ$, we could ask for a procedure enabling one to express a given f in terms of this basis, whose diagonalization is expressed by $\{e(a)Z^i\}_{i=0}^\infty$ being a divided-power sequence over $e(a)$ for a fixed a in F. Note that when F has characteristic 0, then $e(a)Z^i=e(a)\circ(d^i/dx^i)/i!=(d^i/dx^i)_{x=a}/i!$, since all of these have the value $a^{n-i}\binom{n}{i}$ at x^n if $n \ge i$, and 0 if $n<i$. This means that if $g(x)\in F[x]$, then $(e(a)Z^i)(g(x))=g^{(i)}(a)/i!$, the i-th Taylor coefficient of $g(x)$ at $x=a$.

There is also the question of allowing F to be a commutative ring. The reader will have noticed that our examples involved integers, but this does not guarantee that Δf can be diagonalized in terms of f, $Df, \ldots$ with integer coefficients. For example, if $f=(1, 4, 9, 16, \ldots)$, i.e., $f_n=(n+1)^2$ for $n \ge 0$, then its minimal recursive relation is x^3-3x^2+3x-1. The Hankel matrix of f is

$$\begin{bmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{bmatrix},$$

whose inverse has $s_{00}=\frac{31}{28}$. (We are assuming here that characteristic F is zero.) Hence the coefficient of $f\otimes f$ in Δf is not integral. Of course, $x^3-3x^2+3x-1=(x-1)^3$, so that C_f has an "integral" basis of $e(1)$, $e(1)Z$, $e(1)Z^2$. Linearly recursive sequences whose entries are from integral domains or from matrix rings have arisen in linear dynamical systems (e.g., [4]).

Finally, one could try to take ideas concerning linearly recursive sequences and try to interpret them Hopf algebraically. A good exposition of some of the basic properties of linearly recursive sequences can be found in [3]. Conversely, one could try to apply Hopf-theoretic concepts to the further study of linearly recursive sequences. We hope to pursue such ideas in future work.

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